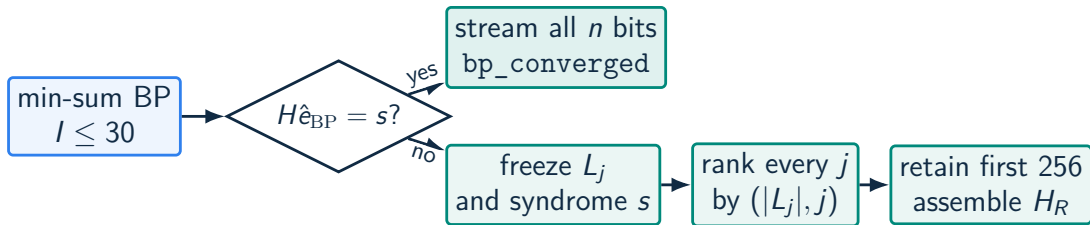


OSD-0 begins where BP stops

BB144 Z-check sector: $m = 936$, $n = 8784$, with 7-bit final BP marginals L_j .



Sparse source

`col_ones[j]` lists touched checks; the full 936×8784 matrix is not runtime state.

Assemble on drain

Ranked j enters slot r ; set $H_R[i, r] = 1$ for every $i \in \text{col_ones}[j]$.

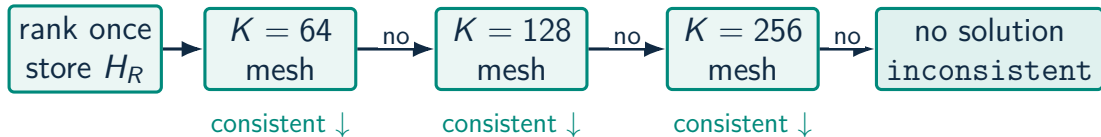
Stored state

$H_R \in \mathbb{F}_2^{936 \times 256}$: 239,616 bits plus 256 original indices.

Current policy: all 8784 columns drain; retain the first $R_{\max} = 256$ (no $|L|$ filter).

One ranking, three solver widths

The $64 \rightarrow 128 \rightarrow 256$ retries reuse the same order and H_R : no BP rerun, no re-sort.



Stream (`selectedIndex[r]`, x_r); every unselected correction bit is zero.

Active-row stream

Solve $H_R[:, 0:K]x_K = s$. Stream only rows with nonzero coefficients or $s_i = 1$; call their count m'_K .

Cycle and area consequences

Sort: 17,568 clocks. Attempt K : $m'_K + 3K + 1$; solved output: K . All three meshes exist, but only one is active.